

数值分析第五次作业

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1. 证明 Legendre 多项式

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

满足递推式:

$$nP_n(x) = (2n-1)xP_{n-1}(x) - (n-1)P_{n-2}(x), \quad n \geq 2.$$

解答: 记 $u_n(x) = (x^2 - 1)^n$, 则

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n = \frac{1}{2^n n!} u_n^{(n)},$$

从而只需证:

$$u_n^{(n)} = 2(2n-1)xu_{n-1}^{(n-1)} - 4(n-1)^2 u_{n-2}^{(n-2)}.$$

在 $(x^2 - 1)u'_n = 2nxu_n$ 两边同求 n 阶导数可得

$$(x^2 - 1)u_n^{(n+1)} + 2nxu_n^{(n)} + n(n-1)u_n^{(n-1)} = 2n^2 u_n^{(n-1)} + 2nxu_n^{(n)},$$

化简可得

$$n(n+1)u_n^{(n-1)} = (x^2 - 1)u_n^{(n+1)}. \quad (1)$$

在 $u_n = (x^2 - 1)u_{n-1}$ 两边同求 n 阶导数, 并结合 (1) 可得

$$\begin{aligned} u_n^{(n)} &= n(n-1)u_{n-1}^{(n-2)} + 2nxu_{n-1}^{(n-1)} + (x^2 - 1)u_{n-1}^{(n)} \\ &= (x^2 - 1)u_{n-1}^{(n)} + 2nxu_{n-1}^{(n-1)} + (x^2 - 1)u_{n-1}^{(n)} \\ &= 2(x^2 - 1)u_{n-1}^{(n)} + 2nxu_{n-1}^{(n-1)}, \end{aligned}$$

故

$$u_{n-1}^{(n)} = \frac{1}{2(x^2 - 1)}(u_n^{(n)} - 2nxu_{n-1}^{(n-1)}). \quad (2)$$

在 $u'_n = 2nxu_{n-1}$ 两边同求 n 阶导数可得

$$u_n^{(n+1)} = 2n^2u_{n-1}^{(n-1)} + 2nxu_{n-1}^{(n)},$$

再代入 (2) 可得

$$\frac{1}{2(x^2 - 1)}(u_{n+1}^{(n+1)} - 2(n+1)xu_n^{(n)}) = 2n^2u_{n-1}^{(n-1)} + \frac{nx}{x^2 - 1}(u_n^{(n)} - 2nxu_{n-1}^{(n-1)}),$$

化简可得

$$u_{n+1}^{(n+1)} = 2(2n+1)xu_n^{(n)} - 4n^2u_{n-1}^{(n-1)},$$

即为

$$u_n^{(n)} = 2(2n-1)xu_{n-1}^{(n-1)} - 4(n-1)^2u_{n-2}^{(n-2)}.$$

□

2. 证明下述多项式为正交多项式, 并推导其递推式.

(1) Chebyshev 多项式: $\omega(x) = (1 - x^2)^{-1/2}$, $[a, b] = [-1, 1]$,

$$P_n(x) = \cos(n \arccos x), \quad |x| \leq 1.$$

(2) Laguerre 多项式: $\omega(x) = e^{-x}$, $[a, b] = [0, +\infty)$,

$$P_n(x) = e^x \frac{d^n}{dx^n}(x^n e^{-x}).$$

(3) Hermite 多项式: $\omega(x) = e^{-x^2}$, $(a, b) = (-\infty, +\infty)$,

$$P_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}(e^{-x^2}).$$

解答:(1) 先证 $P_n(x)$ 为正交多项式, 只需证: $\forall 0 \leq k < n$, 都有

$$\int_{-1}^1 (1 - x^2)^{-1/2} P_k(x) P_n(x) dx = 0.$$

事实上, 令 $x = \cos \theta, \theta \in [0, \pi]$ 可得

$$\int_{-1}^1 (1-x^2)^{-1/2} P_k(x) P_n(x) dx = \int_0^\pi \cos k\theta \cos n\theta d\theta = \frac{1}{2} \int_0^\pi (\cos(k+n)\theta + \cos(k-n)\theta) d\theta = 0.$$

再求递推式. 由于

$$\cos n\theta = 2 \cos \theta \cos(n-1)\theta - \cos(n-2)\theta,$$

再令 $\theta = \arccos x$ 可得

$$P_n(x) = 2xP_{n-1}(x) - P_{n-2}(x).$$

(2) 先证 $P_n(x)$ 为正交多项式, 只需证: $\forall 0 \leq k < n$, 都有

$$\int_0^{+\infty} e^{-x} x^k P_n(x) dx = 0.$$

事实上,

$$\begin{aligned} & \int_0^{+\infty} e^{-x} x^k P_n(x) dx \\ &= \int_0^{+\infty} x^k d\left(\frac{d^{n-1}}{dx^{n-1}}(x^n e^{-x})\right) \\ &= \left(x^k \frac{d^{n-1}}{dx^{n-1}}(x^n e^{-x})\right) \Big|_0^{+\infty} - \int_0^{+\infty} \frac{d^{n-1}}{dx^{n-1}}(x^n e^{-x}) \cdot kx^{k-1} dx \\ &= -k \int_0^{+\infty} x^{k-1} d\left(\frac{d^{n-2}}{dx^{n-2}}(x^n e^{-x})\right) \\ &= \dots \\ &= (-1)^k k! \int_0^{+\infty} d\left(\frac{d^{n-k-1}}{dx^{n-k-1}}(x^n e^{-x})\right) \\ &= (-1)^k k! \left(\frac{d^{n-k-1}}{dx^{n-k-1}}(x^n e^{-x})\right) \Big|_0^{+\infty} = 0, \end{aligned}$$

这里用到了 0 和 $+\infty$ 都是 $x^n e^{-x}$ 的 n 阶零点.

再求递推式. 下面证明:

$$P_n(x) = (2n-1-x)P_{n-1}(x) - (n-1)^2 P_{n-2}(x),$$

再记 $u_n(x) = x^n e^{-x}$, 则只需证:

$$u_n^{(n)} = (2n-1-x)u_{n-1}^{(n-1)} - (n-1)^2 u_{n-2}^{(n-2)}.$$

在 $u_n = xu_{n-1}$ 两边同求 n 阶导数可得

$$u_n^{(n)} = xu_{n-1}^{(n)} + nu_{n-1}^{(n-1)},$$

故

$$u_{n-1}^{(n)} = \frac{1}{x}(u_n^{(n)} - nu_{n-1}^{(n-1)}). \quad (3)$$

在 $u'_n = nu_{n-1} - u_n$ 两边同求 n 阶导数可得

$$u_n^{(n+1)} = nu_{n-1}^{(n)} - u_n^{(n)},$$

再代入 (3) 可得

$$\frac{1}{x}(u_{n+1}^{(n+1)} - (n+1)u_n^{(n)}) = \frac{n}{x}(u_n^{(n)} - nu_{n-1}^{(n-1)}) - u_n^{(n)},$$

化简可得

$$u_{n+1}^{(n+1)} = (2n+1-x)u_n^{(n)} - n^2u_{n-1}^{(n-1)},$$

即为

$$u_n^{(n)} = (2n-1-x)u_{n-1}^{(n-1)} - (n-1)^2u_{n-2}^{(n-2)}.$$

(3) 先证 $P_n(x)$ 为正交多项式, 只需证: $\forall 0 \leq k < n$, 都有

$$\int_{-\infty}^{+\infty} e^{-x^2} x^k P_n(x) dx = 0.$$

事实上,

$$\begin{aligned} & \int_{-\infty}^{+\infty} e^{-x^2} x^k P_n(x) dx \\ &= (-1)^n \int_{-\infty}^{+\infty} x^k d \left(\frac{d^{n-1}}{dx^{n-1}} (e^{-x^2}) \right) \\ &= (-1)^n \left(x^k \frac{d^{n-1}}{dx^{n-1}} (e^{-x^2}) \right) \Big|_{-\infty}^{+\infty} - (-1)^n \int_{-\infty}^{+\infty} \frac{d^{n-1}}{dx^{n-1}} (e^{-x^2}) \cdot kx^{k-1} dx \\ &= (-1)^{n+1} k \int_{-\infty}^{+\infty} x^{k-1} d \left(\frac{d^{n-2}}{dx^{n-2}} (e^{-x^2}) \right) \\ &= \dots \\ &= (-1)^{n+k} k! \int_{-\infty}^{+\infty} d \left(\frac{d^{n-k-1}}{dx^{n-k-1}} (e^{-x^2}) \right) \end{aligned}$$

$$= (-1)^{n+k} k! \left(\frac{d^{n-k-1}}{dx^{n-k-1}} (e^{-x^2}) \right) \Big|_{-\infty}^{+\infty} = 0,$$

这里用到了 $-\infty$ 和 $+\infty$ 都是 e^{-x^2} 的 n 阶零点.

再求递推式. 下面证明:

$$P_n(x) = 2xP_{n-1}(x) - 2(n-1)P_{n-2}(x),$$

再记 $u(x) = e^{-x^2}$, 则只需证:

$$u^{(n)} = -2xu^{(n-1)} - 2(n-1)u^{(n-2)}.$$

在 $u' = -2xu$ 两边同求 n 阶导数可得

$$u^{(n+1)} = -2xu^{(n)} - 2nu^{(n-1)},$$

即为

$$u^{(n)} = -2xu^{(n-1)} - 2(n-1)u^{(n-2)}.$$

□